

Ronit Shahu

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MACHINE LEARNING ENGINEER | DATA SCIENTIST

PROFESSIONAL SUMMARY

AI Scientist and Machine Learning Engineer with over three years of experience taking research concepts and shipping code in production. Built scalable machine learning systems using LangGraph, PyTorch, and production-grade deployment tools, ensuring high uptime for high-throughput inference APIs. Beyond modeling, I engineer robust backend services using FastAPI, Docker, and Kubernetes to manage data-intensive tasks. Prioritize clean, maintainable software design and Responsible AI practices to deliver solutions that are both innovative and trustworthy. Delivered model-driven recommendations improving operational efficiency and decision-speed for business stakeholders.

TECHNICAL SKILLS, MLOps AND PLATFORMS

Machine Learning & GenAI: PyTorch, TensorFlow, Scikit-learn, Deep Learning (CNNs, Transformers), Large Language Models (LLMs), LangGraph, RAG Pipelines, Vector Databases (FAISS, Pinecone), Embeddings, NLP (Summarization, Q&A, NER), Retrieval-Augmented Reasoning, Prompt Engineering, Generative Modeling.

Software Engineering & MLOps: Python, SQL, PySpark, FastAPI, Docker, Kubernetes, CI/CD (GitHub Actions), MLflow, Distributed Data Pipelines, Monitoring & Logging (CloudWatch), Tableau, Power BI, Seaborn/Matplotlib.

Cloud & Data Platforms: Azure, Databricks, AWS, PySpark, REST APIs, Container Orchestration, High-throughput Inference Systems.

Responsible AI & Evaluation: Model Explainability (SHAP), Fairness Analysis, Bias Detection, Compliance-Aligned ML Design, A/B Testing, Cost-Sensitive Optimization, Experimentation Frameworks

WORK EXPERIENCE

Lead Machine Learning Researcher, Canadian Robotics & AI Lab, University of Ottawa, ON | Jan 2025 – May 2025

Invited to lead a cross-disciplinary team supporting a national research effort on AI ethics.

- Worked with a 15-member team to define the project scope, prepare datasets, and build and evaluate models, helping move research ideas into dependable, production-ready systems.
- Created and maintained ML APIs using FastAPI and Docker, deployed on AWS with GitHub Actions, MLflow tracking, and CloudWatch monitoring.
- Introduced a more efficient microservice setup that reduced compute costs by about 50% and made it easier for teams to review evaluation results and collaborate across research areas.

Tech Stack Used: *Python, PyTorch, scikit-learn, FastAPI, Docker, GitHub Actions, AWS EC2, MLflow*

Independent AI Consultant | Oct 2023 – Dec 2025

Working with organizations to design, refine, and deploy AI solutions that support operational needs.

- Built ML, NLP, and GenAI models using PyTorch, TensorFlow, and Spark/PySpark, helping teams improve model accuracy by 25% and cut data processing time by 35% across varied datasets. Translated analytical findings into executive-ready recommendations.
- Developed and deployed AI pipelines on Azure ML and Databricks, using MLflow and modular Python code to reduce training and deployment cycles by 50% and lower inference latency by 30% for real-time and batch use.
- Created retrieval-based and graph-structured LLM pipelines using LangGraph, improving automation coverage by 40% and improving retrieval accuracy by 20% for decision-making tasks.

Tech Stack Used: *Python, TensorFlow, LangGraph, Azure ML/Databricks, Docker, SQL, Spark/PySpark*

Software Developer – Life science and Health Care, Persistent Systems Limited, India | June 2022 – July 2023

Worked on healthcare-focused AI solutions with cross-functional global teams.

- Built custom spaCy NER models for cancer-related terms and deployed them through Azure ML endpoints, reducing data preparation time and improving insight accuracy by 40%.
- Maintained and enhanced Flask/FastAPI microservices, containerized with Docker and deployed through Azure ML

and CI/CD pipelines, increasing information retrieval speed by 12%. Partnered with business stakeholders to identify revenue-impacting use cases.

- Collaborated with data scientists, DevOps engineers, and clinical SMEs to turn healthcare data challenges into deployable ML applications that enforce fairness constraints using bias-correction algorithms.
- Worked with multi-modal datasets including structured transactions, text corpora, and document embeddings for retrieval workflows. Presented insights in PowerPoint/Tableau dashboards

Tech Stack Used: *Python, spaCy, TensorFlow, FastAPI, Azure ML, Databricks, CI/CD, SQL, PySpark*

Software Developer Intern (Co-op) – Persistent Systems Limited, India | Dec 2021 – June 2022

Contributed to ML and automation projects supporting internal compliance teams.

- Automated image and PDF verification using machine learning and OCR through Azure ML pipelines, cutting manual review effort by 70% and reducing operational costs by 35%.
- Built NLP and ML models with spaCy, TensorFlow, and scikit-learn, improving prediction accuracy and speeding up data processing.
- Designed modular Python ETL and feature-engineering pipelines using SQL, Spark, and PySpark for large-scale, real-world datasets, and helped deploy distributed ML workflows on Azure ML. Queried multi-million-row datasets via SQL to aggregate behavioral patterns.

Tech Stack Used: *Python, TensorFlow, scikit-learn, spaCy, OCR, Azure ML, Databricks, Docker, SQL, Spark, PySpark*

EDUCATION

Master of System Science and Engineering (Interdisciplinary AI, Dept. of Computer Science) University of Ottawa, ON	2023 – 2025
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- **Cumulative GPA:** 8.9/10.0 (3.8/4.0 Equivalent). **Relevant Coursework:** Foundations of Machine Learning, Data Science Applications, Systems Optimization & Management, Ethics for AI & Robotics (Responsible AI), Systems Engineering, Modelling & Simulation.

Bachelor of Engineering in Electronics Engineering Shri Ramdeobaba College of Engineering and Management, India	2018 – 2022
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AWARDS AND PUBLICATIONS

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| • Research Fellowship Scholarship, University of Ottawa, ON | 2025 |
| • Top Talent 2023, Persistent Systems (for measurable ML project impact), Persistent Systems Limited, India | 2023 |
| • 2nd Runner-Up, Global AI Hackathon Semicolons 2023, India | 2023 |
| • <u>Audio + Text based Emotion Recognition System (Published: Springer LNEE)</u> | 2021 |
| • <u>Detection of COVID-19 using Machine Learning & Android App (Published: NeuroQuantology)</u> | 2021 |

KEY PROJECTS

Hybrid Fraud Detection + GenAI Triage Pipeline

- Developed an end-to-end fraud detection system using Logistic Regression, Random Forest, LightGBM, and a PyTorch MLP, trained on 6M+ Paysim transactions and optimized for low-latency inference across multiple model families.
- Built an agentic fraud-triage workflow with Google Gemini, where the LLM calls structured tools to compare model outputs, resolve disagreements, retrieve similar past transactions via a TF-IDF RAG index, and generate grounded APPROVE / REVIEW / DECLINE recommendations with clear business rationale.
- Implemented production-grade MLOps workflows including reproducible training scripts, model serialization, standardized artifacts, and MLflow-ready tracking to support future deployment as a FastAPI + Docker microservice.
- Created a transparent, auditable reasoning layer where the LLM integrates multi-model scores + retrieved patterns, producing human-readable explanations that reduce analyst review time and improve trust in the system.

End-to-End Credit Risk Scoring Pipeline (Responsible AI Focus)

- Built a credit scoring engine focusing on Responsible AI compliance; utilized SHAP (Global & Local) to provide transparent explainability for every credit decision. Conducted rigorous fairness audits across sensitive groups (e.g., gender, status) using Weight of Evidence (WOE) analysis to eliminate algorithmic bias.
- Designed a cost-sensitive evaluation framework that optimized decision thresholds based on the monetary impact of False Negatives vs. False Positives, aligning model performance with business P&L.